MIL-S-51294
1 July 1968
(See section 6)

MILITARY SPECIFICATION

STANNIC CHLORIDE, ANHYDROUS, ANALYZED REAGENT

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers reagent grade anhydrous stannic chloride (hereinafter referred to as stannic chloride).

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specifiction to the extent specified herein.

SPECIFICATIONS

FEDERAL

TT-W-572 - Wood-Preservative; Water-Repellent.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636 - Box, Fiberboard.
PPP-C-300 - Chemicals, Liquid; Packaging and Packing of.

MILITARY

MIL-P-15011 - Pallets, Material Handling, Wood, Post Construction, 4-Way Entry.

STANDARDS

MILITARY

MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-147 - Palletized and Containerized Unit Loads, 40" x 48" 4-Way (Partial) Pallet Skids, Runners, or Pallet-Type Base.

FSC 6810

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(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

CODE OF FEDERAL REGULATIONS

49 CFR 171-179 - Department of Transportation Rules and Regulations for the Transportation of Explosives and Other Dangerous Articles.

(The Department of Transportation regulations are a part of the Code of Federal Regulations available from the Superintendent of Documents, Government Printing Office, Washington, D. C. 20402. Orders for the above publication should cite "49 CFR 171-179.")

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM) STANDARDS

D1193-66 - Reagent Water.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Appearance. Stannic chloride shall be a colorless fuming liquid.

3.2 Chemical characteristics. Stannic chloride shall conform to the chemical characteristics of table I when tested as specified therein.

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Table I. Chemical characteristics

Characteristic	Percent by weight		Test paragraph
	Minimum	Maximum	
Assay (as $SnCl_4$)	99.0	----	4.2.4.1
Solubility	To pass test	----	4.2.4.2
Free chlorine (Cl)	----	To pass test	4.2.4.3
Sulfate (SO_4)	----	0.001	4.2.4.4
Substances not precipitated by H_2S (as sulfates)	----	0.05	4.2.4.5
Iron (Fe)	----	0.001	4.2.4.6
Arsenic (As)	----	0.0005	4.2.4.7

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.

4.1.1 Supplier's responsibility. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to specified requirements.

4.1.2 Objective evidence. The supplier shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1 and section 5 for which specific inspection has not been provided in this specification have been satisfied.

4.2 Quality conformance inspection.

4.2.1 Lotting. A lot shall consist of the stannic chloride produced by one manufacturer, at one plant, from the same materials, and under essentially the same manufacturing conditions provided the operation is continuous. In the event the process is a batch operation, each batch shall constitute a lot (see 6.3).

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4.2.2 Sampling for test. Sampling shall be conducted as specified in table II. Approximately equal portions of stannic chloride shall be taken from each container in the sample, composited, and placed in a suitable clean, dry container labeled to identify the lot and containers from which it was taken. The total composited sample shall be no less than 1/4 pound.

Table II. Sampling for test

Number of containers in lot or batch	Number of containers in sample
2-25	2
26-150	3
151-1,200	5
1,201-7,000	8
7,001-20,000	10
over 20,000	20

4.2.3 Inspection procedure. The composited sample of stannic chloride shall be tested as specified in 4.2.4. Each test shall be conducted in duplicate analysis. Failure of either analysis of any test shall be cause for rejection.

4.2.4 Tests. Water in accordance with ASTM D1193 and reagent grade chemicals shall be used throughout the tests. Where applicable, blank determinations shall be run and corrections applied where significant. Tests shall be conducted as follows (see 6.4):

4.2.4.1 Assay (as $SnCl_4$). Weigh to the nearest 0.1 milligram (mg) approximately 0.55 gram (g) of the specimen and dissolve in 10 milliliters (ml) of 1 to 1 hydrochloric acid in a 400-ml beaker. Add 25.0 ml of 0.1M ethylenediamine tetraacetic acid (EDTA) and heat to near boiling for 2 to 3 minutes. Add 1 or 2 drops of thymol blue indicator and then add hexamethylenetetramine until the solution turns yellow. Add a quantity of hexamethylenetetramine equal to the amount previously added and wash down the sides of the beaker with water. Heat for 2 to 3 minutes, cool to room temperature, and dilute the solution to 250 ml with water. Add 1 g of ammonium nitrate and a small quantity of methyl thymol blue indicator (mixed 1 part per 100 parts of potassium nitrate). Slowly back titrate with 0.1M lead nitrate solution, with continuous stirring, to a blue or green end point that persists for 30 seconds. Calculate percent stannic chloride as follows:

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$$\text{Percent stannic chloride} = \frac{26\ (25A-BC)}{W}$$

where: A = Molarity of EDTA solution used,
B = Molarity of lead nitrate solution used,
C = Milliliters of lead nitrate solution used, and
W = Weight of specimen in grams.

4.2.4.2 Solubility. Weigh 5.0 g of the specimen and carefully dissolve in 100 ml of water. The solution shall be clear and colorless.

4.2.4.3 Free chlorine. Weigh 2.28 g of the specimen and dissolve in 20 ml of water. Cool to room temperature, add 1 ml of chloroform and 2 drops of 10-percent potassium iodide solution, and shake. No violet color shall appear in the chloroform layer within 1 minute.

4.2.4.4 Sulfate. Weigh 25.0 g of the specimen, carefully dissolve in 175 ml of water, and neutralize with ammonium hydroxide solution using litmus paper as indicator. Dilute to 250 ml with water and allow the precipitate to settle. Decant 200 ml of the supernatant liquid through a filter, neutralize the filtrate to litmus paper with concentrated hydrochloric acid and add 2 ml excess concentrated hydrochloric acid. Heat the filtrate solution to boiling, add 5 ml of 12-percent barium chloride solution, and allow to stand overnight. If any precipitate is formed, filter, wash thoroughly with hot water, ignite in a tared platinum crucible weighed to the nearest 0.1 mg, cool to room temperature, and weigh to the nearest 0.1 mg. Calculate the percent sulfate as follows:

$$\text{Percent sulfate} = \frac{41.2\ (A-B)}{W}$$

where: A = Weight of ignited precipitate in grams,
B = Weight of ignited precipitate in a blank test in grams, and
W = Weight of specimen present in 200-ml aliquot in grams.

4.2.4.5 Substances not precipitated by hydrogen sulfide. Weigh 4.558 g of the specimen and dilute with 220 ml of water, heat to 70° $\pm$ 2° C, precipitate the tin completely with hydrogen sulfide, and filter. To 100 ml of the filtrate, add 0.5 ml of concentrated sulfuric acid, evaporate to dryness in a tared porcelain crucible weighed to the nearest 0.1 mg, ignite, cool to room temperature, and weigh to the nearest 0.1 mg (retain the residue for 4.2.4.6). The weight of the crucible and residue minus the weight of the crucible shall not exceed 1.0 mg.

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4.2.4.6 Iron. To the residue from 4.2.4.5, add 2 ml of concentrated hydrochloric acid and 0.5 ml of concentrated nitric acid and slowly evaporate to dryness on a steam bath. Take up the residue with 2 ml of concentrated hydrochloric acid and dilute to 50 ml with water. Add 30 to 50 mg of ammonium persulfate crystals and 3 ml of 30-percent ammonium thiocyanate solution and mix. Any red color produced shall be no darker than that produced in a blank to which 0.02 mg of iron (Fe) ion has been added.

4.2.4.7 Arsenic. Test 0.400 g of the specimen for arsenic by the Gutzeit method (see 6.5). The stain produced shall be no darker than that produced in a separate test in which 0.002 mg of arsenic (As) is substituted for the specimen and tested in the same manner.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A. One-quarter pound ± 0.05 ounce avoirdupois of stannic chloride shall be packaged in a type I, class 1 bottle utilizing inclosure b (bottles in metal cans) as specified in PPP-C-300. Closure of bottle shall be leak-proof, and shall be either a glass stopper or a high impact resistant, buttress thread, plastic screw cap. The glass stopper shall be secured in placed by corrosion resistant wire. The plastic screw cap shall neither affect nor be affected by stannic chloride.

5.1.1.1 Intermediate packaging. Twelve bottles of stannic chloride, packaged as specified in 5.1.1, shall be intermediately packaged in accordance with the applicable requirements of PPP-C-300 except that table IV therein shall not be used. The bottles shall be arranged in a three bottle by four bottle pattern. Fiberboard separators shall not be required.

5.1.2 Level C. Unless otherwise specified, a 1/4-pound quantity of stannic chloride shall be packaged to provide adequate protection against contamination, deterioration, and damage from the supply source to the first receiving activity for immediate use (see 6.2).

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A. Four boxes of stannic chloride, intermediately packaged as specified in 5.1.1.1, shall be packed for shipment in a nailed wood box conforming to class 2, style 4-1/2, for a type 2 (average) load of PPP-B-621. All inner and outer surfaces of each wood box shall be coated with wood preservative by immersion for 3 minutes in material conforming to type II, composition A of TT-W-572. The box shall be closed and strapped using galvanized strapping in accordance with the applicable requirements specified in the appendix of PPP-B-621.

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5.2.2 Level B. Four boxes of stannic chloride, intermediately packaged as specified in 5.1.1.1, shall be packed for shipment in a fiberboard box conforming to style RSC, grade V3c of PPP-B-636. Closure shall be by taping in accordance with the requirements for taping style RSC boxes to be shipped as exterior containers and then reinforced by banding as specified in the appendix of PPP-B-636.

5.2.3 Level C. Stannic chloride, packaged as specified in 5.1.2, shall be packed to insure carrier acceptance and safe delivery to the first domestic destination. Containers shall be in compliance with Department of Transportation regulations or the regulations of other common carriers applicable to the mode of transportation.

5.3 Marking. In addition to the special marking for unit containers specified in 5.3.1 and any special marking required by the contract or order, unit and intermediate packages and shipping containers shall be marked in accordance with Department of Transportation regulations and MIL-STD-129 (see 6.2).

5.3.1 Special marking. Each unit container shall be durably and legibly labeled to show the following special information:

(a) Empirical formula.
(b) Formula weight.
(c) Manufacturer's lot number.
(d) Assay.
(e) Analysis or maximum limits of impurities.
(f) Precautionary marking:

Avoid skin contact.
Fumes irritating.
Avoid breathing of vapors.

5.4 Palletization. When specified in the contract or order, shipping containers of 5.2 shall be palletized in accordance with the applicable requirements of MIL-STD-147 utilizing the hardwood, 4-way, 40- by 48-inch, double-wing pallet of MIL-P-15011 (see 6.2).

6. NOTES

6.1 Intended use. Stannic chloride is intended for use as a general laboratory reagent.

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6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Level of packaging and packing required.
(c) Quantity other than 1/4 pound for level C packaging, if required.
(d) Special marking other than 5.3.1, if required.
(e) Palletization, if required.

6.3 Batch. A batch is defined as that quantity of material which has been manufactured by some unit chemical process and subjected to some physical mixing operation intended to make the final product substantially uniform.

6.4 Test precaution. Care should be taken whenever stannic chloride is diluted with water because of the heat produced. Stannic chloride is a caustic liquid and should be kept well stoppered in a tightly closed bottle or container because of its irritiating fumes.

6.5 Gutzeit method for arsenic. A description of the apparatus and procedure for determining arsenic by the Gutzeit method may be found under General Procedures in Reagent Chemicals, American Chemical Society Specifications, 1960 edition, or in other standard analytical reference books.

6.6 Significant places. For the purpose of determining conformance with this specification, an observed or calculated value shall be rounded off "to the nearest unit" in the last right-hand place of figures used in expressing the limiting value, in accordance with the rounding-off method of the Recommended Practices for Designating Significant Places in Specified Limiting Values (ASTM E29).

6.7 Supersession data. This specification includes the requirements of Edgewood Arsenal Purchase Description 4-214, dated 5 December 1952.

Custodians:

Army - MU
Air Force - 68

Review activities:

Army - MD, MU(FA)
Air Force - 68
DSA - GS

User activity:

Army - SM

Preparing activity:

Army - MU(EA)

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